Week 3: Statistics

What is Statistics?

Statistics is the science of collecting, organising, compiling, computing, summarising, analysing, and interpreting data for the purpose of decision-making and passing information to the public.

In the context of modelling and simulation, statistics is essential because:

- Real-world data feeds into models as input.
- Simulation outputs are themselves statistical (especially in stochastic models).
- We need statistical tools to compare scenarios, draw conclusions, and quantify uncertainty.

Uses of Statistics

1. **Planning and budgeting** — governments and businesses use statistics to project demand, allocate resources, and forecast revenue.
2. **Assessing the variation of a system** — measuring how much output fluctuates around its average tells us whether the system is stable.
3. **Studying associations between two objects or systems** — does class attendance correlate with exam scores? Does fuel price affect transport demand?
4. **Disseminating research findings** — research becomes trustworthy when supported by statistical evidence.

Measures of Central Tendency

Mean, median, and mode are the three classical *measures of central tendency*. They describe the **centre** or **typical value** of a data set — answering the question, *"What is a representative single number for this data?"*

Ungrouped Data

Worked Example

Given the data set: **1, 2, 2, 3, 4, 5, 2**

Mean (Arithmetic Average)

Add all values and divide by the count:

$$\text{Mean} = \frac{1+2+3+2+4+5+2}{7} = \frac{19}{7} \approx 2.714$$

Median (Middle Value)

First, arrange in ascending order: **1, 2, 2, 2, 3, 4, 5**

The middle value (the 4th of 7 values) is **2**.

$$\text{Median} = 2$$

> If the data set has an *even* number of values, the median is the average of the two middle values.

Mode (Most Frequent Value)

The number 2 appears three times — more than any other. So:

$$\text{Mode} = 2$$

> A data set can have **no mode** (all values appear once), **one mode** (unimodal), or **multiple modes** (bimodal, multimodal).

When to Use Each Measure

Measure	Best For	Example
Mean	Normal, symmetric numerical data without extreme outliers.	Average exam score in a class.
Median	Data with extreme values (outliers) or skewed distributions.	Average income (since billionaires distort the mean).
Mode	Categorical data, or finding the most common category.	Most popular shoe size sold in a shop.

> **Real-world example of why this matters:** If Aliko Dangote walks into a room of 99 average Nigerians, the *mean* net worth jumps to millions of dollars — meaningless for describing a "typical" person in the room. The *median* barely moves, giving a more honest summary.

Grouped Data: Mean, Median, and Mode

When data is presented in a frequency distribution (grouped into class intervals), the formulas change because individual values are no longer known — only the class each value falls into.

Example Frequency Distribution: Height Data

Height	Frequency ()	Midpoint (x)	x	Cumulative Frequency (CF)
1 – 3	3	2	6	3
4 – 6	4	5	20	7
7 – 9	7	8	56	14
10 – 12	18	11	198	32
13 – 15	6	14	84	38
16 – 18	10	17	170	48
19 – 21	2	20	40	50
Total	$\sum = 50$		$\sum x = 574$	

The **midpoint** is the average of the lower and upper class limits (e.g., for class 1–3: midpoint = (1+3)/2 = 2). We treat all values in a class as if they sit at the midpoint.

Mean of Grouped Data

$$\bar{x} = \frac{\sum x}{\sum} = \frac{574}{50} = 11.48$$

Median Formula (Grouped)

$$\text{Median} = L_m + \left[\frac{\frac{n}{2} - F}{m}\right] \times$$

Where:

- L_m = lower class boundary of the median class
- n = total frequency
- F = cumulative frequency *before* the median class
- $_m$ = frequency of the median class
- = class width

The **median class** is the class containing the $(n/2)^{\text{th}}$ observation. Here, $n/2 = 25$, and the cumulative frequency just reaches/exceeds 25 in the class **10–12** (CF goes from 14 to 32). So the median class is 10–12.

Plug in: $L_m = 10$, $n = 50$, $F = 14$, ${}_m = 18$, $= 3$:

$$\text{Median} = 10 + \left[\frac{25 - 14}{18}\right] \times 3 = 10 + \frac{11}{18} \times 3 = 10 + 1.833 = 11.83$$

Mode Formula (Grouped)

$$\text{Mode} = L_m + \left[\frac{{}_1 - {}_0}{2{}_1 - {}_0 - {}_2}\right] \times$$

Where:

- L_m = lower class boundary of the modal class (class with highest frequency)
- ${}_1$ = frequency of the modal class
- ${}_0$ = frequency of the class *before* the modal class
- ${}_2$ = frequency of the class *after* the modal class
- = class width

The **modal class** is 10–12 (frequency 18, the highest). So $L_m = 10$, ${}_1 = 18$, ${}_0 = 7$, ${}_2 = 6$, $= 3$:

$$\text{Mode} = 10 + \left[\frac{18 - 7}{2(18) - 7 - 6}\right] \times 3 = 10 + \frac{11}{23} \times 3 = 10 + 1.435 = 11.43$$

Variance and Standard Deviation

Where central tendency tells us about the *centre* of the data, **variance** and **standard deviation** tell us about the **spread** — how far data points typically lie from the mean.

Why Spread Matters

Two data sets can have the same mean but behave very differently.

> **Example:** Two students both have an average of 70 over five exams.
>
> - Student A scored: 70, 70, 70, 70, 70 (very consistent).
> - Student B scored: 50, 60, 70, 80, 90 (highly variable).
>
> The mean alone hides this difference. The standard deviation reveals it.

Definitions

- **Variance (^{2})** — the *average of squared differences from the mean*. Squaring eliminates negative signs and emphasises large deviations.

- **Standard Deviation ()** — the *square root of variance*. Taking the square root returns the value to the original units (so a variance of 25 marks² becomes a standard deviation of 5 marks).

Interpretation

- **Low variance / standard deviation** → data clusters tightly around the mean (consistent, predictable).
- **High variance / standard deviation** → data is widely spread (volatile, unpredictable).

We typically **report standard deviation** because it shares the units of the data and is easier to interpret. **Variance** is more useful for further statistical analysis (e.g., ANOVA, regression).

Formula for Grouped Data

$$^2 = \frac{\sum x^2}{\sum} - \left[\frac{\sum x}{\sum}\right]^2$$

$$= {}^2$$

> This is the *population* variance formula. It is essentially $E[X^2] - (E[X])^2$ — the expected value of the square minus the square of the expected value.

Worked Example 1: Student Performance

The following table shows student performance in an examination. Find the mean, variance, and standard deviation.

Marks		x (midpoint)	x^2	x^2	x
30 – 34	3	32	1,024	3,072	96
35 – 39	6	37	1,369	8,214	222
40 – 44	13	42	1,764	22,932	546
45 – 49	18	47	2,209	39,762	846
50 – 54	15	52	2,704	40,560	780
55 – 59	7	57	3,249	22,743	399
60 – 64	4	62	3,844	15,376	248
65 – 69	2	67	4,489	8,978	134
Total	**68**			**161,637**	**3,271**

Mean

$$\bar{x} = \frac{\sum x}{\sum} = \frac{3{,}271}{68} \approx 48.10$$

Variance

$$^2 = \frac{\sum x^2}{\sum} - \left[\frac{\sum x}{\sum}\right]^2 = \frac{161{,}637}{68} - (48.10)^2$$

$$^2 = 2{,}377.01 - 2{,}313.89 = 63.12$$

Standard Deviation

$$= 63.12 \approx 7.94$$

Interpretation: Students scored on average about 48 marks, and most fall within roughly ±8 marks of that average. About 68% of students (under a normal-distribution assumption) scored between 40 and 56.

Worked Example 2: Class Intervals

Class		x (midpoint)	x	x^2	x^2
1 – 10	5	5.5	27.5	30.25	151.25
11 – 20	8	15.5	124	240.25	1,922
21 – 30	12	25.5	306	650.25	7,803
31 – 40	7	35.5	248.5	1,260.25	8,821.75
41 – 50	3	45.5	136.5	2,070.25	6,210.75
Total	**35**		**842.5**		**24,908.75**

Mean

$$\bar{x} = \frac{842.5}{35} \approx 24.07$$

Variance

$$^2 = \frac{24{,}908.75}{35} - (24.07)^2 = 711.68 - 579.43 = 132.25$$

Standard Deviation

$$= 132.25 \approx 11.50$$

Interpretation: This data set has a higher standard deviation (11.5) than the student-marks data (7.94), meaning it is *more spread out* relative to its centre.

Correlation Coefficient

The **correlation coefficient** is a single number that tells you **how strongly two variables are related** and **in what direction**. The most common version is **Pearson's correlation coefficient**, which measures the *linear* relationship between two variables.

Range and Meaning

The coefficient always falls in the range $[-1, +1]$:

Value	Meaning
$= +1$	**Perfect positive correlation** — both variables increase together exactly in step.
$0 < \quad < 1$	Positive correlation — they tend to increase together.
$= 0$	**No linear relationship** — knowing one tells you nothing linear about the other.
$-1 < \quad < 0$	Negative correlation — when one increases, the other tends to decrease.
$= -1$	**Perfect negative correlation** — exact inverse relationship.

Rough Strength Guide

	Strength
0.0 – 0.2	Very weak / none
0.2 – 0.4	Weak
0.4 – 0.6	Moderate
0.6 – 0.8	Strong
0.8 – 1.0	Very strong

> **Important caution:** Correlation is **not causation**. Ice-cream sales and drowning rates are positively correlated, but ice cream does not cause drowning — both increase in summer. This is why correlation must be interpreted carefully.

Formula

$$= \frac{n\sum xy - \sum x \sum y}{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}$$

Where n is the number of paired observations.

Worked Example 1: Calculating Correlation

x	y	xy	x^2	y^2
5	70	350	25	4,900
7	62	434	49	3,844
8	63	504	64	3,969
10	65	650	100	4,225
12	45	540	144	2,025
15	50	750	225	2,500
17	22	374	289	484
18	48	864	324	2,304
20	40	800	400	1,600
23	25	575	529	625
$\sum x = 135$	$\sum y = 490$	$\sum xy = 5{,}841$	$\sum x^2 = 2{,}149$	$\sum y^2 = 26{,}476$

With $n = 10$:

$$= \frac{10(5{,}841) - (135)(490)}{[10(2{,}149) - 135^2][10(26{,}476) - 490^2]}$$

Numerator:

$$10(5{,}841) - (135)(490) = 58{,}410 - 66{,}150 = -7{,}740$$

Denominator:

$$10(2{,}149) - 18{,}225 = 21{,}490 - 18{,}225 = 3{,}265$$

$$10(26{,}476) - 240{,}100 = 264{,}760 - 240{,}100 = 24{,}660$$

$$3{,}265 \times 24{,}660 = 80{,}514{,}900 \approx 8{,}973.01$$

Result:

$$= \frac{-7{,}740}{8{,}973.01} \approx -0.8626$$

Interpretation: ≈ -0.86 indicates a **strong negative correlation** — as x increases, y tends to decrease in a fairly consistent linear way.

Worked Example 2: Hours Studied vs. Exam Grade

Let x represent hours studied and y represent exam grade.

x	y	xy	x^2	y^2
1	40	40	1	1,600
2	50	100	4	2,500
3	60	180	9	3,600
4	70	280	16	4,900
5	80	400	25	6,400
$\sum x = 15$	$\sum y = 300$	$\sum xy = 1{,}000$	$\sum x^2 = 55$	$\sum y^2 = 19{,}000$

With $n = 5$:

$$= \frac{5(1{,}000) - (15)(300)}{[5(55) - 15^2][5(19{,}000) - 300^2]}$$

$$= \frac{5{,}000 - 4{,}500}{(275 - 225)(95{,}000 - 90{,}000)} = \frac{500}{50 \times 5{,}000} = \frac{500}{250{,}000} = \frac{500}{500} = 1$$

Interpretation: $= 1$ — a **perfect positive correlation**. Every extra hour of study adds exactly 10 marks. (Real exam data is never this clean — this is a textbook scenario.)

Alternative Form: Using Deviations from the Mean

The same correlation can also be computed using deviations from the mean:

$$= \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2 \cdot \sum(y - \bar{y})^2}$$

First find the means:

$$\bar{x} = \frac{15}{5} = 3, \quad \bar{y} = \frac{300}{5} = 60$$

Then build the deviation table:

$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
-2	-20	4	400	40
-1	-10	1	100	10
0	0	0	0	0
1	10	1	100	10
2	20	4	400	40
0	**0**	**10**	**1,000**	**100**

(Note: the deviation columns always sum to zero — a useful check.)

Apply the formula:

$$= \frac{100}{10 \times 1{,}000} = \frac{100}{10{,}000} = \frac{100}{100} = 1$$

Same answer ✓ — the two formulas are mathematically equivalent.

Why Statistics Matters for Modelling and Simulation

Bringing this back to the course theme:

- When **collecting input data** for a simulation, you compute means and standard deviations to describe the distributions feeding your model.
- When **analysing simulation output** (especially from stochastic models), you need statistics to summarise multiple runs and assess uncertainty.
- **Correlation** helps detect relationships between variables you may want to include in or exclude from your model.
- **Variance** is central to assessing whether a system is stable or volatile — a key question in queueing, manufacturing, finance, and reliability engineering.

In short: you cannot model what you cannot measure, and statistics gives us the language for measurement.